

Saejoon Park

Fairfax, VA | (571)-373-7778 | park.p.saejoon@gmail.com | [Portfolio](#) | US Citizen

EDUCATION

Cornell University Ithaca, New York

Aug 2024 - May 2028

Bachelor of Science in Electrical and Computer Engineering (GPA: 3.6/4.0)

SKILLS

Programming Languages: Python, C, C++, Swift, Java, Verilog

Software/Platforms: GitHub, KiCad, Quartus, LTSpice, Fusion360, Claude Code, Claude Design, Figma MCP

Languages: English (fluent), Korean (Conversational)

PROJECTS

Computer Vision + Glove-based Tracking Computer Mouse | Ithaca, NY

Jan 2025

- ATmega32U4; custom conductive inputs; IMU; assembled a position tracking glove device (Valorant compatible)
- Computer Vision (OpenCV) allows for hand gesture tracking(simulated clicks/volume control) at 30 fps

Thermal Electric Generator Powered Mouse | Fairfax, VA

Aug 2023

- Patent Pending hand thermal energy-powered computer mouse
- Implemented TEG Modules and a transformer to convert a 20mV output from TEG to 3.3V

Steel Electric Bike from Scratch | Burke, VA

May 2025 - Aug 2025

- Custom-cut and assembled steel chassis; 1000W motor; 50V accumulator

Audio Signal Decoding Circuit & Filter Design | Ithaca, NY

April 2026

- Designed and built a 6th-order MFB bandpass filter to extract encoded audio secrets.
- Conducted automated frequency response sweeps and eliminated multi-volt DC offsets safely.

Sensory Overload Wearable | Ithaca, NY

Feb 2025 - Now

- Developing a custom PCB for a lap wearable, accommodating individuals with sensory processing disorder

Political AI Camera | Ithaca, NY

Nov 2025 - Now

- Developed the Raspberry Pi-based hardware featuring a 12.1 MP sensor and software using the Picamera2 library
- Calls OpenAI API with the gpt-image-1 model, prompted to output different political perspectives of given image

Oddity 1 by IBX, Chrome Extension | Ithaca, NY

Feb 2026 - March 2026

- Using chenglou/pretext JS library, GDocs API, and HEX color tracking system, created a strategy for on-screen GDocs annotations (previously not possible w/o Google Whitelist), solving <canvas> rendering difficulties.
- Developed a "Cursor for Writing" system in GDocs using a clean-room rewritten Claude Code opensource code

ACTIVITIES

Cornell Assistive Technologies Project Team | Ithaca, NY

Feb 2025 - Apr 2026

- Project-based engineering group for developing assistive technologies for people with disabilities

Cornell Machine Poetics Research, Sang Leigh | Ithaca, NY

Nov 2025 - Apr 2026

- Explores how expressive technology and AI can fit into products and systems to enhance human experience

Cornell Extended Reality Project Team | Ithaca, NY

Sept 2025 - Jan 2026

- Developed Custom Haptic Gloves for a Virtual Reality System

WORK EXPERIENCE

Co-Founder/COO of IBX LLC

Mar 2025 - Now

- Currently Frontend Developer/Outreach: working with FCPS CTIO and Director of Counseling for app approval

HONORS

USPTO Provisional Patent for TEG Hot Swap Computer Mouse

Fall 2023